LRSR-ADMM-Net: A Joint Low-Rank and Sparse Recovery Network for SAR Imaging

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***Abstract*— Synthetic aperture radar (SAR) imaging with sub-Nyquist sampled echo is a challenging task. Compressed sensing (CS) has been widely applied in this case to reconstruct the unambiguous image. The CS-based methods need to set the iterative parameters manually, but the appropriate parameters are usually difficult to obtain. Besides, such methods require a large number of iterations to obtain satisfactory results, which seriously restricts their practical applications. Moreover, the observation scene of SAR is not sparse in some cases. In this article, we aim at proposing an efficient and effective imaging method for nonsparse observation scenes with reduced data. First, considering the characteristics of nonsparse observation scenes in SAR imaging, we model the SAR imaging problem as a joint low-rank and sparse matrices recovery problem. After that, the iterative alternating direction method of multipliers (ADMMs) to solve the above problem is unrolled into a layer-fixed deep neural network with trainable parameters, in which the learnable parameters are layer-varied. The threshold parameters, as well as the weight parameter between the sparse part and low-rank part of each layer, are learned adaptively instead of manually tuned. Experiments prove that the proposed low-rank and sparse recovery LRSR-ADMM-Net is capable of reconstructing the nonsparse observed scene with high efficiency and precision. Particularly, the proposed LRSR-ADMM-Net yields better reconstruction performance while maintaining high computational efficiency compared with the state-of-the-art iterative recovery methods and the trainable sparse-based network methods.**

***Index Terms*— Alternating direction method of multipliers (ADMMs), deep unfolding, imaging method, sub-Nyquist, synthetic aperture radar (SAR).**

I. Introduction

AS AN active microwave imaging system, synthetic aperture radar (SAR) has broad application prospects in the fields of Earth mapping, disaster monitoring, reconnaissance, and surveillance because of its all-day and all-weather imaging capability of ground and maritime targets [1], [2]. In recent years, the spatial resolution of the SAR system has continuously been improved with the expansion of practical applications [3]–[5]. Higher sampling rates in both azimuth and range are required with the finer spatial resolution. As a result, it poses big challenges to the radar sampling device. Moreover, the transmission, storage, and processing of the enormous data sampled by the SAR system are also intractable problems, especially for small unmanned aerial vehicle (UAV) platforms.

It has been shown that compressive sensing (CS) techniques can recover the original signal from highly incomplete samples or measurements than that required by Shannon–Nyquist sampling theory when the signal is sparse or sparse in some basis [6]–[9]. In [10]–[15], the authors have applied the CS technique to SAR imaging where the observation model was constructed through the inverse process of the computationally efficient imaging algorithm, and the unambiguous imaging problem with the subsampled data was modeled as sparse recovery problem. The imaging results show that unambiguous image of sparse scene can be obtained where the number of samples can be greatly reduced. However, the CS-based imaging methods above are suitable only for sparse imaging scene and cannot be directly applied to nonsparse observation scene. In addition to CS, matrix sensing (MS) theory was also applied to SAR imaging [16], [17]. MS utilizes the inertial correlation of matrix, i.e., low-rank property, to reconstruct the original image [18], [19]. In comparison with the CS-based method, the MS-based method is more effective for nonsparse observation scene because the inertial structure of the matrix is maintained. However, the standard MS-based method faces performance degradation when the imaging scene exists anomaly dominant scatterers, such as buildings. In these cases, the low-rank assumption of the imaging scene is no more valid. Moreover, the sparsity and low-rankness of the SAR image were jointly utilized in [20]–[22] to recover the nonsparse SAR image. The reconstruction performance shows the superiority of the joint sparse and low-rank recovery model. However, the above iterative reconstruction method needs to set the iterative parameters manually, but the appropriate parameters are usually difficult to obtain. Besides, such method requires a large amount of iterations to obtain satisfactory results, which seriously restricts its prospects in practical applications.

On the other hand, as a kind of data-driven method, the deep learning (DL) technique can learn intrinsic features from the existing data through training, so that the knowledge can be

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used in the subsequent processing. The handcraft modeling and parameter selection in the model-driven technique can be avoided. Therefore, the DL technique can obtain better performance compared with the traditional model-driven technique. Deep neural networks (DNNs) have demonstrated their availability in many applications, including image processing, computer vision, and natural language processing. DL techniques also bring new inspiration to radar imaging. The potential of applying DNN to radar imaging was explored in [23] where a complex-valued convolutional neural network (CNN) was proposed. In addition, there are some research works focusing on generating high-resolution radar images from low-resolution images via CNNs [24]–[27]. These methods provide fast feed-forward inferences, but for lack of principled knowledge may fail to give the desired results when the input deviates from the mode of training data distribution. This situation becomes more serious in SAR imaging due to its variable observation geometry.

Another idea aims to find an interpretable solution, dubbed deep unfolding [28], [29], which introduces learnable architectures into the conventional iterative recovery methods. In the unfolding networks, the basic structure is derived from the iterative updating process of the applied algorithm, and the learnable parameters in the network are automatically obtained through training. Therefore, the unfolding network can be regarded as a combination of model-driven and data-driven techniques, which can inherit the advantages of both techniques. In recent years, several unfolding networks have been proposed and they focus on different application scenarios. In [30], alternating direction method of multipliers (ADMMs)-CSNet was proposed for magnetic resonance imaging (MRI) and optical image reconstruction, and the results show that ADMM-CSNet can achieve favorable performance with high computational efficiency. Several networks were applied to inverse SAR (ISAR) imaging, including AF-AMPNet [31] and pattern-coupled sparse Bayesian learning (PCSBL)-generalized approximate message passing (GAMP)-Net [32]. The unfolding TPSSI-Net [33], LFIST-Net [34], and RMIST-Net [35] for 3-D millimeter-wave (mmW) holography were proposed. Meanwhile, there are some works using unfolding or unrolled deep networks for SAR imaging serving different purposes. For example, the target-enhancing matched filter (MF)-ADMM-Net [36] highlights the targets from the background clutters, the sparsity-cognizant total least-square (STLS)-linearized alternating direction method of multipliers (LADMM)-Net [37] can perform motion compensation while imaging by simultaneously estimating reconstructed image and phase error, and networks proposed in [38]–[41] aim at SAR imaging of sparse scenes. These techniques have been widely applied in ISAR imaging, 3-D mmW holography, and sparse scene imaging, but their applications in SAR nonsparse scene imaging have not received enough attention.

In this article, an imaging method for a nonsparse observation scenes with reduced data is proposed. First, the nonsparse scene imaging problem is modeled as a joint low-rank and sparse matrix recovery problem by fully utilizing the properties of nonsparse observation scenes. Following that, the iterative alternating ADMM to solve the above problem is unfolded into a layer-fixed DNN with trainable parameters. Note that the convergence of the network is guaranteed by constructing the learnable parameters expressions with monotonicity. The threshold parameters and the weight parameter between the sparse part and low-rank part of each layer are learned automatically rather than handcrafting adjustment. The proposed LRSR-ADMM-Net is capable of reconstructing the nonsparse observed scene with high efficiency and precision. Moreover, the proposed LRSR-ADMM-Net outperforms the state-of-the-art iterative recovery methods and the trainable sparse-based network methods.

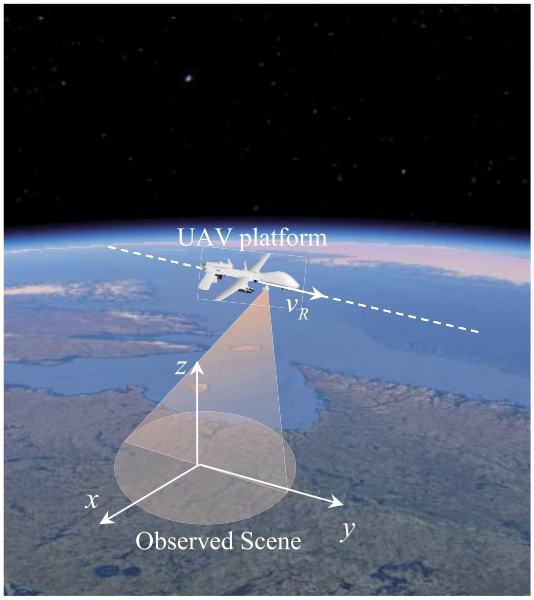

Fig. 1. Observation geometry of SAR imaging.

The rest of this article is organized as follows. Section II establishes the observation geometry and imaging model for SAR nonsparse scene imaging. The iterative recovery method for SAR imaging is given in Section III. The proposed LRSR-ADMM-Net is presented in Section IV. Simulation results and discussion are provided in Section V to verify the effectiveness of the proposed method. Conclusions are given in Section VI.

II. SAR Observation Geometry and Imaging Model

The observation geometry of SAR imaging is given in Fig. 1. The radar is mounted on a UAV or airplane, and the flight path of the airborne platform is parallel to the y axis at a constant velocity of v_R. Meanwhile, the radar illuminates the observed area and receives the reflected echo from the target where linear frequency-modulated (LFM) pulses are transmitted at a constant rate.

Suppose the imaging area of SAR is rectangular, with M pixels in range and N pixels in azimuth. The imaging area is denoted by $\mathbf{X}$, which is an $M \times N$ matrix and whose (m, n) entry is $\mathbf{X}(m, n)$. Meanwhile, the echo received by the radar is denoted as $\mathbf{Y} \in \mathbb{C}^{P \times Q}$ where P and Q indicate the total sample number of the azimuth time and range time, respectively. The received signal at the pth azimuth pulse and qth range sequence is denoted as $\mathbf{Y}(p, q)$. The relation between the echo $\mathbf{Y}(p, q)$ and reflection coefficients $\mathbf{X}$ is given as follows:

$$\mathbf{Y}(p,q) = \sum_{m=1}^{M} \sum_{n=1}^{N} \mathbf{X}(m,n) \varphi(p,q,m,n) \tag{1}$$

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where

$$\begin{aligned}\varphi(p,q,m,n) &= \omega_r\left(\tau_q - \frac{2R(p,m,n)}{c}\right)\omega_a\left(\frac{p}{T_a}\right)\\ &\quad\times\exp\left\{-j4\pi\frac{R(p,m,n)}{\lambda}\right\}\\ &\quad\times\exp\left\{j\pi K_r\left(\tau_q - \frac{2R(p,m,n)}{c}\right)^2\right\}. \quad (2)\end{aligned}$$

Note that $\omega_r(\cdot)$ and $\omega_a(\cdot)$ are the range and azimuth envelope functions, respectively. Furthermore, $R(p,m,n)$ is the instant range between the target located at position (m,n) at azimuth pulse p, λ represents the wavelength of the transmitted signal, c is the speed of the light, K_r is the rate of the transmitted LFM signal, and τ_q is the range sequence at qth sampling.

It can be seen from (2) that the calculation of $\varphi(p,q,m,n)$ is time consuming because there are nearly $PQ \times MN$ elements that need to be calculated separately. In order to establish the echo collection model in a more concise and efficient way, the inverse process of the effective SAR imaging algorithms [10], [11], [42], such as the Range-Doppler algorithm and the Chirp Scaling algorithm, was applied.

The process of the Chirp Scaling algorithm which can establish the mapping from the echo $\mathbf{Y}$ to the final SAR image $\mathbf{X}$ is written as follows:

$$\begin{aligned}\mathbf{X} &= \Im(\mathbf{Y})\\ &= \mathfrak{F}_a^{-1}\left\{\mathfrak{P}_3\left\{\mathfrak{F}_r^{-1}\left\{\mathfrak{P}_2\left\{\mathfrak{F}_r\{\mathfrak{P}_1\{\mathfrak{F}_a\{\mathbf{Y}\}\}\}\right\}\right\}\right\}\right\} \quad (3)\end{aligned}$$

where $\mathfrak{F}_r$ and $\mathfrak{F}_a$ represent range and azimuth Fourier transform, respectively, $\mathfrak{F}_r^{-1}$ and $\mathfrak{F}_a^{-1}$ represent range and azimuth inverse Fourier transform, respectively, $\mathfrak{P}_1$, $\mathfrak{P}_2$ and $\mathfrak{P}_3$ are the phase multiplication processing of chirp-scaling, range compression, and azimuth compression, respectively.

To establish the mapping from the imaging scene to its corresponding echo, the decoupling model $\Re(\cdot)$ is applied, which is given in (4)

$$\begin{aligned}\mathbf{Y} &= \Re(\mathbf{X})\\ &= \mathfrak{F}_a^{-1}\left\{(\mathfrak{P}_1)^*\left\{\mathfrak{F}_r^{-1}\left\{(\mathfrak{P}_2)^*\left\{\mathfrak{F}_r\left\{(\mathfrak{P}_3)^*\{\mathfrak{F}_a\{\mathbf{X}\}\}\right\}\right\}\right\}\right\}\right\}. \quad (4)\end{aligned}$$

In (4), $(\cdot)^{-1}$ represents the inverse process, and $(\cdot)^*$ is the conjugate operation. Note that $\Re(\cdot)$ is computational efficient since the calculation and storage of the high-dimensional matrix in (2) are avoided. The dimension of the decoupling model is almost the same as that of the observed scene.

The subsampling of the echo in azimuth and range can be regarded as the operation in that a reduced set of azimuth and range elements in echo matrix $\mathbf{Y}$ is taken. Note that the reduced set is randomly taken in this article. $\mathbf{\Theta}_a$ and $\mathbf{\Theta}_r$ are denoted as the subsampling matrices along azimuth and range directions, respectively. After subsampling, the reduced echo received by the receiver is given as

$$\mathbf{Y}_d = \mathbf{\Theta}_a\mathbf{Y}\mathbf{\Theta}_r = \mathbf{\Theta}_a\Re(\mathbf{X})\mathbf{\Theta}_r. \quad (5)$$

In CS theory, the signal $\mathrm{vec}(\mathbf{X})$ can be recovered from the highly incomplete samples by minimizing the following l_0-norm problem when the signal is sparse, namely:

$$\begin{aligned}&\arg\min_{\mathrm{vec}(\mathbf{X})} \|\mathrm{vec}(\mathbf{X})\|_0\\ &\text{s.t. } \mathbf{Y}_d = \mathbf{\Theta}_a\Re(\mathbf{X})\mathbf{\Theta}_r. \quad (6)\end{aligned}$$

In CS theory, the previous problem can be solved by greedy strategies [43], or l_1-norm minimization [44]. It should be noted that the problem in (6) aims at obtaining a sparse solution for the imaging scene. To solve the problem of nonsparse imaging scene, some sparse transformations which can obtain the sparse representation of the imaging scene are applied [45], [46]. However, the correlation information of the imaging scene is ignored by simply processing the imaging scene as a sparse or compressible vector.

In [17], to exploit the 2-D correlation structure of the imaging scene, the authors applied low-rank property to model $\mathbf{X}$. $\mathbf{X}$ can be recovered by solving the following rank minimization problem, namely:

$$\begin{aligned}&\arg\min_{\mathbf{X}} \mathrm{rank}(\mathbf{X})\\ &\text{s.t. } \mathbf{Y}_d = \mathbf{\Theta}_a\Re(\mathbf{X})\mathbf{\Theta}_r. \quad (7)\end{aligned}$$

Similarly, the problem in (7) can be solved using greedy algorithms [47] or nuclear norm minimization [48]. In practical applications, the standard MS-based imaging algorithms suffer from performance degradation when the imaging scene exits anomaly dominant scatterers, such as buildings. In these cases, the low-rank assumption of the imaging scene is not valid.

In [20] and [21], instead of using sparse property or low-rank property of the SAR imaging scene separately, the sparsity and low-rankness of the SAR image were jointly analyzed. It has been found that the SAR images $\mathbf{X}$ can be modeled as the sum of a low-rank matrix $\mathbf{L}$ and a sparse matrix $\mathbf{S}$ consisting of anomaly dominant scatterers, that is

$$\mathbf{X} = \mathbf{L} + \mathbf{S}. \quad (8)$$

After that the nonsparse scene imaging problem is modeled as a joint rank and l_0-norm minimization problem

$$\begin{aligned}&\arg\min_{\mathbf{L},\mathbf{S}} \mathrm{rank}(\mathbf{L}) + \lambda\|\mathbf{S}\|_0\\ &\text{s.t. } \mathbf{Y}_d = \mathbf{\Theta}_a\Re(\mathbf{L}+\mathbf{S})\mathbf{\Theta}_r. \quad (9)\end{aligned}$$

III. Iterative Joint Low-Rank and Sparse Recovery Method for SAR Imaging

In this section, we briefly introduce the recovery method for the joint minimization problem in (9) to lay the foundation of the following network design.

Since the joint recovery problem in (9) is an NP-hard problem, the l_0-norm minimization problem and rank minimization problem in (9) are first converted into the l_1-norm minimization problem and the nuclear norm minimization problem, respectively, which is given by

$$\begin{aligned}&\arg\min_{\mathbf{L},\mathbf{S}} \|\mathbf{L}\|_* + \lambda\|\mathbf{S}\|_1\\ &\text{s.t. } \mathbf{Y}_d = \mathbf{\Theta}_a\Re(\mathbf{L}+\mathbf{S})\mathbf{\Theta}_r \quad (10)\end{aligned}$$

where $\|\cdot\|_*$ represents the nuclear norm of a matrix, which is the sum of its singular values.

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To solve the equality constrained optimization problem, the augmented Lagrangian function of (10) is

$$\begin{aligned} L_\rho(\mathbf{L},\mathbf{S},\mathbf{Q},\rho) &= \min_{\mathbf{L},\mathbf{S}} \|\mathbf{L}\|_* + \lambda\|\mathbf{S}\|_1 + \langle \mathbf{Q}, \Theta_a\Re(\mathbf{L}+\mathbf{S})\Theta_r - \mathbf{Y}_d\rangle \\ &\quad + \frac{\rho}{2}\|\Theta_a\Re(\mathbf{L}+\mathbf{S})\Theta_r - \mathbf{Y}_d\|_F^2 \end{aligned} \tag{11}$$

where $\mathbf{Q}$ is the Lagrange multipliers, $\langle\cdot,\cdot\rangle$ is the inner product, and ρ is the penalty parameter.

In order to adapt the decoupling model, a modified ADMM was proposed in [21]. The ADMM decomposes the minimization of L_ρ with respect to $(\mathbf{L},\mathbf{S})$ into two subproblems which minimize $\mathbf{L}$ and $\mathbf{S}$, respectively. The iterative process of the modified ADMM method for joint low-rank and sparse recovery is given in detail as follows:

A. Update Low-Rank Part $\boldsymbol{L}$

The updating scheme of the low-rank part is given in (12) which is shown at the bottom of the page.

In (12), $\eta = \|\Theta_a\Re(\mathbf{I})\|_2^2$. The updated low-rank part can be obtained by applying proximal operator of the nuclear norm.

B. Update Sparse Part $\boldsymbol{S}$

The updating scheme of the sparse part is given in (13) which is shown at the bottom of the page.

The updated sparse part can also be obtained by applying soft-thresholding operator.

C. Update Lagrange Multipliers $\boldsymbol{Q}$

The updating scheme of Lagrange multipliers is given as follows:

$$\mathbf{Q}^{k+1} = \mathbf{Q}^k + \rho^k\left(\Theta_a\Re\left(\mathbf{L}^{k+1}+\mathbf{S}^{k+1}\right)\Theta_r - \mathbf{Y}_d\right). \tag{14}$$

D. Update Penalty Parameter ρ

The adaptive updating strategy for the penalty parameter is

$$\rho^{k+1} = \min(\alpha\rho^k, \rho_{\max}) \tag{15}$$

where $\rho_{\max}$ is an upper bound of ρ, and $\alpha \geq 1$ is a constant.

In the iterative recovery method, if the update rate of both the low-rank and sparse parts is less than a predefined value, the iteration is stopped. Finally, the reconstruction of the imaging scene is achieved.

From the iterative process of the ADMM method, several updating parameters need to be predetermined, including the weight parameter between the sparse part and low-rank part, the threshold parameters for the update of the low-rank and sparse parts, and the step size of the penalty parameter. However, the values of these parameters are hard to be appropriately determined. Moreover, this method requires a large amount of iterations to obtain satisfactory results, which seriously restricts its practical applications.

IV. Proposed LRSR-ADMM-Net for SAR Nonsparse Scene Imaging

In this section, the framework of the proposed LRSR-ADMM-Net is derived based on the unrolling process of the previous joint low-rank and sparse recovery method. Specifically, each layer of the proposed LRSR-ADMM-Net consists of three modules, i.e., the low-rank part updating module, the sparse part updating module, and the Lagrange multiplier updating module. The architecture of the proposed LRSR-ADMM-Net is shown in Fig. 2. Moreover, the detailed implementation of the update process in the LRSR-ADMM-Net is provided, including the design of the learnable step size and threshold parameters to ensure rapid convergence.

The details of the proposed LRSR-ADMM-Net are provided as follows:

1) Low-Rank Part Updating Module(LM^t): Recalling the update process of the low-rank part in the ADMM method as shown in (12), it first needs to calculate the intermediate variable $\mathbf{A}^k$ in the current step, which is given as

$$\mathbf{A}^k = \Im\left(\Theta_a^{\mathbf{T}}\left\{\frac{\mathbf{Q}^k}{\rho^k} + \left(\Theta_a\Re\left(\mathbf{L}^k+\mathbf{S}^k\right)\Theta_r - \mathbf{Y}_d\right)\right\}\Theta_r^{\mathbf{T}}\right)\Big/\eta. \tag{16}$$

After that, the update of the low-rank part is given as follows:

$$\mathbf{L}^{k+1} = D_{1/\rho^k\eta}\left(\mathbf{L}^k - \mathbf{A}^k\right) = \mathbf{U}M_{1/\rho^k\eta}\left(\sum\right)\mathbf{V}^*. \tag{17}$$

Note that $\mathbf{X} = \mathbf{U}\sum\mathbf{V}^*$ and $\sum = \mathrm{diag}(\{\sigma_i\}_{1\leq i\leq r})$ is the singular value decomposition (SVD) of $\mathbf{X}$ whose rank is r. Denote $D_\tau(\cdot)$ as the proximal operator of the nuclear norm which is given as

$$D_\tau(\mathbf{X}) = \mathbf{U}M_\tau\left(\sum\right)\mathbf{V}^* \tag{18}$$

where

$$M_\tau\left(\sum\right) = \mathrm{diag}\left((\sigma_i-\tau)_+\right). \tag{19}$$

Note that $x_+ = \max(0,x)$.

From the detailed updating process, it can be found that the threshold parameter $1/\rho^k\eta$ is utilized to control the value of the low-rank part in the iteration. However, the proper value of $1/\rho^k\eta$ is hard to predetermine. Moreover, the value of the threshold parameter should be accordingly adjusted to adapt

$$\mathbf{L}^{k+1} = \arg\min_{\mathbf{L}} \|\mathbf{L}\|_* + \frac{\rho^k\eta}{2}\left\|\mathbf{L} - \mathbf{L}^k + \Im\left(\Theta_a^{\mathbf{T}}\left\{\frac{\mathbf{Q}^k}{\rho^k} + \left(\Theta_a\Re\left(\mathbf{L}^k+\mathbf{S}^k\right)\Theta_r - \mathbf{Y}_d\right)\right\}\Theta_r^{\mathbf{T}}\right)\Big/\eta\right\|_F^2 \tag{12}$$

$$\mathbf{S}^{k+1} = \arg\min_{\mathbf{S}} \lambda\|\mathbf{S}\|_1 + \frac{\rho^k\eta}{2}\left\|\mathbf{S} - \mathbf{S}^k + \Im\left(\Theta_a^{\mathbf{T}}\left\{\frac{\mathbf{Q}^k}{\rho^k} + \left(\Theta_a\Re\left(\mathbf{L}^{k+1}+\mathbf{S}^k\right)\Theta_r - \mathbf{Y}_d\right)\right\}\Theta_r^{\mathbf{T}}\right)\Big/\eta\right\|_F^2 \tag{13}$$

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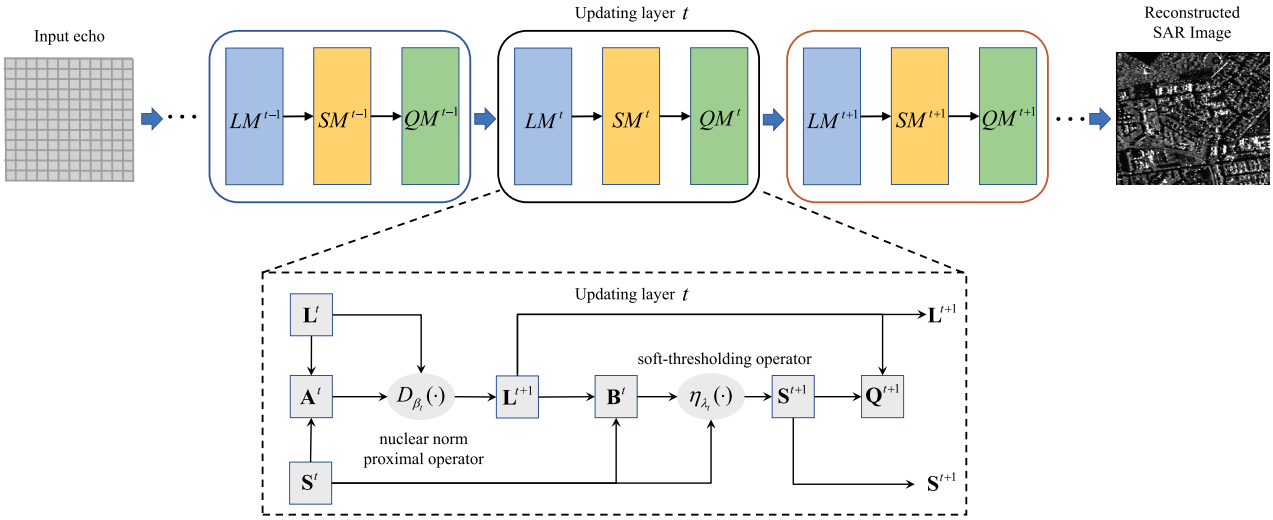

Fig. 2. Architecture of the proposed LRSR-ADMM-Net from subsampled echo to the final reconstructed image.

to different observed scenes in various practical applications of SAR.

The threshold parameter for updating the low-rank part in the LRSR-ADMM-Net is set as a learnable parameter β_t where the subscript t represents the tth layer. Besides, according to the ADMM method, the value of the threshold parameter should decrease step by step to guarantee convergence. Therefore, the learnable parameter β_t for the update of the low-rank part is designed as

$$\beta_t = \omega_{1,t} a_{1,t}^{1-t}. \tag{20}$$

The according adjustment of β_t can be achieved by tuning the layer-varied learnable parameters $\omega_{1,t}$ and $a_{1,t}$. Note that the threshold parameter for updating the low-rank part in ADMM is monotonic decreasing, thus the value of learnable parameter $a_{1,t}$ should be larger than 1.

Meanwhile, from the process of the proximal operator of the nuclear norm, its threshold operation on singular value is the same as the Rectified Linear Unit (ReLU) function. Therefore, the proximal operator of the nuclear norm in (17) can be written as

$$\mathbf{L}^{t+1} = D_{\beta_t}\left(\mathbf{L}^t - \mathbf{A}^t\right) = \mathbf{U}\,\mathrm{diag}(\mathrm{ReLU}(\sigma_i - \beta_t))\mathbf{V}^*. \tag{21}$$

Note that the variable $\mathbf{A}^t$ in (21) is essentially the same as $\mathbf{A}^k$ in (16), with the superscript t denoting the tth layer of the network and the superscript k denoting the kth iteration in the iterative recovery method. The calculation of these two variables are basically the same as well, except the representation of the parameter (i.e., $\mathbf{Q}^t, \mathbf{L}^t, \mathbf{S}^t, \ldots$ for $\mathbf{A}^t$ and $\mathbf{Q}^k, \mathbf{L}^k, \mathbf{S}^k, \ldots$ for $\mathbf{A}^k$).

2) Sparse Part Updating Module (SM^t): Similarly, to update the sparse part in the ADMM as shown in (13), it first needs to calculate the intermediate variable $\mathbf{B}^k$ in the current step, which is given as

$$\mathbf{B}^k = \Im\left(\Theta_a^{\mathbf{T}}\left\{\frac{\mathbf{Q}^k}{\rho^k} + \left(\Theta_a \Re\left(\mathbf{L}^{k+1} + \mathbf{S}^k\right)\Theta_r - \mathbf{Y}_d\right)\right\}\Theta_r^{\mathbf{T}}\right) \Big/ \eta. \tag{22}$$

The update of the sparse part by applying soft-thresholding operator in the ADMM is given as follows:

$$\mathbf{S}^{k+1} = \eta_{\lambda/\rho^k \eta}(\mathbf{S}^k - \mathbf{B}^k) \tag{23}$$

where $\eta_\delta(\mathbf{X})$ represents the soft-thresholding operator, which is given as

$$\eta_\delta(\mathbf{X}) = \mathrm{sign}(\mathbf{X})(|\mathbf{X}| - \delta)_+. \tag{24}$$

Note that

$$\mathrm{sign}(\mathbf{X}) = \begin{cases} \mathbf{X}/|\mathbf{X}|, & \text{for } \mathbf{X} \neq \mathbf{0} \\ 0, & \text{for } \mathbf{X} = \mathbf{0}. \end{cases} \tag{25}$$

It can be found that the updated value of the sparse part in each iteration is related to the value of the threshold parameter $\lambda/\rho^k\eta$. Meanwhile, the update process is also controlled by the weight parameter λ between the low-rank part and the sparse part. Likewise, the values of these two parameters $1/\rho^k$ and λ are hard to be appropriately determined due to the lack of prior information. In particular, the weight parameter λ should be accordingly adjusted for different observed scenes because the anomaly scatters contribute the majority in some observation scenes while the correlative scatters are dominant in others.

Instead of setting the threshold parameter $\lambda/\rho^k\eta$ manually, a learnable parameter λ_t varying with respect to the layer is introduced in the LRSR-ADMM-Net, which is denoted as

$$\lambda_t = \omega_{2,t} a_{2,t}^{1-t}. \tag{26}$$

Note that λ_t should also be decreased with respect to the process of iteration in the ADMM according to its convergence analysis. Therefore, the value of learnable parameter $a_{2,t}$ also should be larger than 1.

From the expression of the soft-thresholding operator, it achieves the same function as the ReLU function. Therefore, the update of the sparse part in the LRSR-ADMM-Net can be achieved by

$$\mathbf{S}^{t+1} = \eta_{\lambda_t}(\mathbf{S}^t - \mathbf{B}^t) = \frac{\mathbf{S}^t - \mathbf{B}^t}{|\mathbf{S}^t - \mathbf{B}^t|}\,\mathrm{ReLU}(\left|\mathbf{S}^t - \mathbf{B}^t\right| - \lambda_t). \tag{27}$$

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Similarly, the variable $\mathbf{B}^t$ in (27) is essentially the same as $\mathbf{B}^k$ in (22), except the representation of the parameter (i.e., $\mathbf{Q}^t, \mathbf{L}^{t+1}, \mathbf{S}^t, \ldots$ for $\mathbf{B}^t$ and $\mathbf{Q}^k, \mathbf{L}^{k+1}, \mathbf{S}^k, \ldots$ for $\mathbf{B}^k$).

3) Lagrange Multiplier Updating Module(QM^t): Lagrange multiplier is applied to the next step of the update of the low-rank and sparse parts. Recalling the update process of the Lagrange multiplier as shown in (14) in the ADMM, the penalty parameter ρ^k is utilized to control the update speed. The value of the penalty parameter ρ^k in each iteration is calculated by (15), which needs to predetermine the initial value, step size α, and upper bound. Note that the penalty parameter significantly influences the convergence speed and reconstruction performance. However, how to choose the proper value is a challenging task in the conventional ADMM.

In the LRSR-ADMM-Net, the penalty parameter which is used for the update of the Lagrange multiplier is set as a layer-varied learnable parameter α_t, which is given as

$$\alpha_t = \omega_{3,t} a_{3,t}^{t-1}. \tag{28}$$

Note that the learnable parameter $a_{3,t}$ should be a positive parameter larger than 1 considering the monotonic increasing requirement of the penalty parameter α_t.

Therefore, the update of the Lagrange multiplier in the LRSR-ADMM-Net is given as

$$\mathbf{Q}^{t+1} = \mathbf{Q}^t + \alpha_t \left(\Theta_a \Re\left(\mathbf{L}^{t+1} + \mathbf{S}^{t+1}\right)\Theta_r - \mathbf{Y}_d\right). \tag{29}$$

4) Loss Function: Suppose that $\{\mathbf{X}_i, \mathbf{Y}_{d,i}\}, i = 1, \ldots, N_{\text{train}}$ is the training set. The trainable parameter set is denoted as $\mathbf{W} = \{\omega_{1,t}, \omega_{2,t}, \omega_{3,t}, a_{1,t}, a_{2,t}, a_{3,t}\}$. The training of the LRSR-ADMM-Net parameters can be achieved by minimizing the following loss function:

$$\text{Loss}(\mathbf{W}) = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} \frac{\left\|\mathbf{X}_i - \hat{\mathbf{X}}_i\right\|_F^2}{\|\mathbf{X}_i\|_F^2} \tag{30}$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm, $\hat{\mathbf{X}}_i$ is the output of the LRSR-ADMM-Net corresponding to the input SAR image $\mathbf{X}_i$, N_{train} is the number of training samples.

The parameters of LRSR-ADMM-Net can be optimized via backpropagation with a variety of optimizers [49], such as stochastic gradient descent (SGD), L-Broyden-Fletcher-Goldfarb-Shanno (BFGS), and Adam.

The architectures of the LRSR-ADMM-Net are derived from the iterative ADMM and the optimal parameters can be obtained through the learning from the data. Therefore, the difficulties in manually setting parameters in conventional ADMM can be avoided. Besides, the convergence speed and reconstruction performance can be greatly improved.

There are some state-of-the-art unrolled deep networks for ISAR imaging, 3-D holography and SAR imaging of sparse scenes, such as AF-AMPNet [31], PCSBL-GAMP-Net [32], LFIST-Net [34], and SAR sparse scenes imaging networks [40], [41]. It should be noted that they are essentially unfolding networks dedicated to sparse recovery problems, i.e., only including sparse updating module. The LRSR-ADMM-Net proposed in this article adopts the network architecture of simultaneous updating low-rank and sparse modules. Therefore, the sparsity and low-rankness of the nonsparse observation scene can be jointly utilized at the same time, which results in better reconstruction performance.

TABLE I
SYSTEM PARAMETERS OF EXPERIMENT 1

Parameter	Value
Carrier frequency	10 GHz
Signal bandwidth	300 MHz
Pulse width	1 μs
Fast time sampling rate	450 MHz
Synthetic aperture time	2.32 s
Velocity of the platform	120 m/s
Slant range	4 km

Fig. 3. Six randomly selected images from training dataset in experiment 1.

V. NUMERICAL RESULTS AND DISCUSSIONS

In this section, the performance of the proposed imaging method is assessed using simulated data. The iterative recovery algorithms, including the sparse recovery method [13], the low-rank recovery method [17], the joint sparse and low-rank recovery method [21], and the state-of-the-art unrolled deep network, i.e., the LFIST-Net [34], are adopted for comparisons.

In the simulations, the proposed LRSR-ADMM-Net is implemented using the Pytorch framework on Windows 10 operating system. The network is trained using Adam optimizer with a 3.80-GHz Advanced Micro Devices (AMD) Ryzen 7 5800X central processing unit (CPU) and an NVIDIA RTX 3060Ti graphics processing unit (GPU). The proposed networks are trained with the following hyperparameters: the learning rate is 0.005, batch size is set to 15, the number of updating layers is 6, and the optimization steps before early stop when the valid loss does not descend any longer are 800.

A. Experiment 1

The system parameters of the SAR system used in experiment 1 are provided in Table I where the transmitted signal bandwidth is 300 MHz. It is assumed that the radar platform flies at a speed of 120 m/s. Considering the lack of real-measured training data, the training dataset is generated via synthetic method in this experiment. The training set consists of 300 SAR images which are cropped from the original image by Airbus. Note that the size of the cropped image is 512 × 512. Fig. 3 shows some typical images which are

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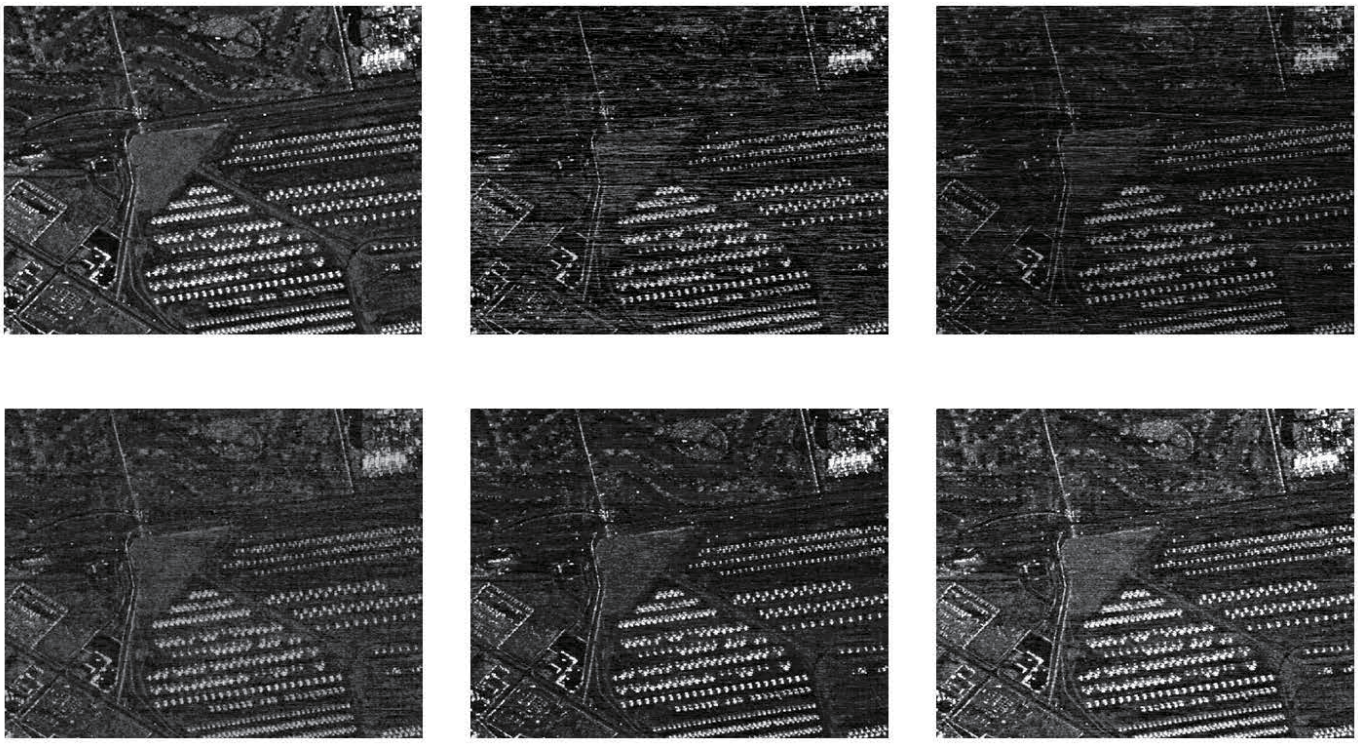

Fig. 4. Imaging results of scene 3 in experiment 1 obtained by different methods with 60% sampling rate. Columns 1 to 3 of the first row correspond to the original SAR image, sparse recovery method, and LFIST-Net, respectively. Columns 1 to 3 of the second row correspond to low-rank recovery method, joint low-rank, and sparse recovery method and the proposed method, respectively.

TABLE II
PROCESSING TIME COMPARISONS OF DIFFERENT METHODS

CS-based method	LFIST-Net	MS-based method	Joint recovery method	LRSR-ADMM-Net
1.23 s	0.72 s	23.85 s	30.56 s	0.98 s

randomly selected from the training dataset. It can be observed that the scenes contain complex surface features, including residential areas, roads, woods, and buildings. Meanwhile, we can also find that these imaging scenes are not sparse in general. The corresponding echoes of the scenes are generated with the amplitude of the image in the dataset using the system parameters given in Table I.

We compare the reconstruction performance of different methods with the subsampled echo. The echo is subsampled in azimuth randomly with a sampling rate of 60%. The ground-truth SAR image of scene 3 for testing is shown in Fig. 4. The imaging results obtained by the sparse recovery method, i.e., the real-time sparse SAR imaging method [13], MS-based method [17], joint sparse and low-rank recovery method [21], the state-of-the-art unrolled deep network, i.e., LFIST-Net [34], and the proposed method are also shown in Fig. 4. It can be observed that the imaging results recovered by the CS-based method and the LFIST-Net are aliased seriously because the original scene is not sparse. Therefore, the state-of-the-art LFIST-Net suffers performance degradation when the nonsparse scene is to be recovered. Note that the reconstruction performance of the recovered image by the MS-based method improves compared with the sparse-based methods because the correlation of the nonsparse imaging scene is utilized. However, the recovered SAR image obtained by the MS-based method is a bit blurry, especially for the strong scatterers. This is because the low-rank assumption is not accurate for SAR images with strong scatterers. Furthermore, the methods based on the joint low-rank and sparse recovery, i.e., the iterative joint low-rank and sparse recovery method and the proposed LRSR-ADMM-Net achieve significant improvement in reconstruction performance. However, a large number of iterations is required to obtain the final results for the joint low-rank and sparse recovery method while the computation time required by the proposed LRSR-ADMM-Net is significantly reduced because only six updating layer network is applied in this simulation. The processing time comparisons of different methods are given in Table II. It can be found the computation time of the conventional joint low-rank and sparse recovery method is 30.56 s because hundreds of iterations are required while the computation time of the proposed LRSR-ADMM-Net is only 0.98 s, which shows the superiority of the proposed method in computation efficiency.

The imaging results of another imaging scene in the testing dataset, i.e., scene 25, are provided in Fig. 5 to further illustrate the reconstruction performance. Similarly, better

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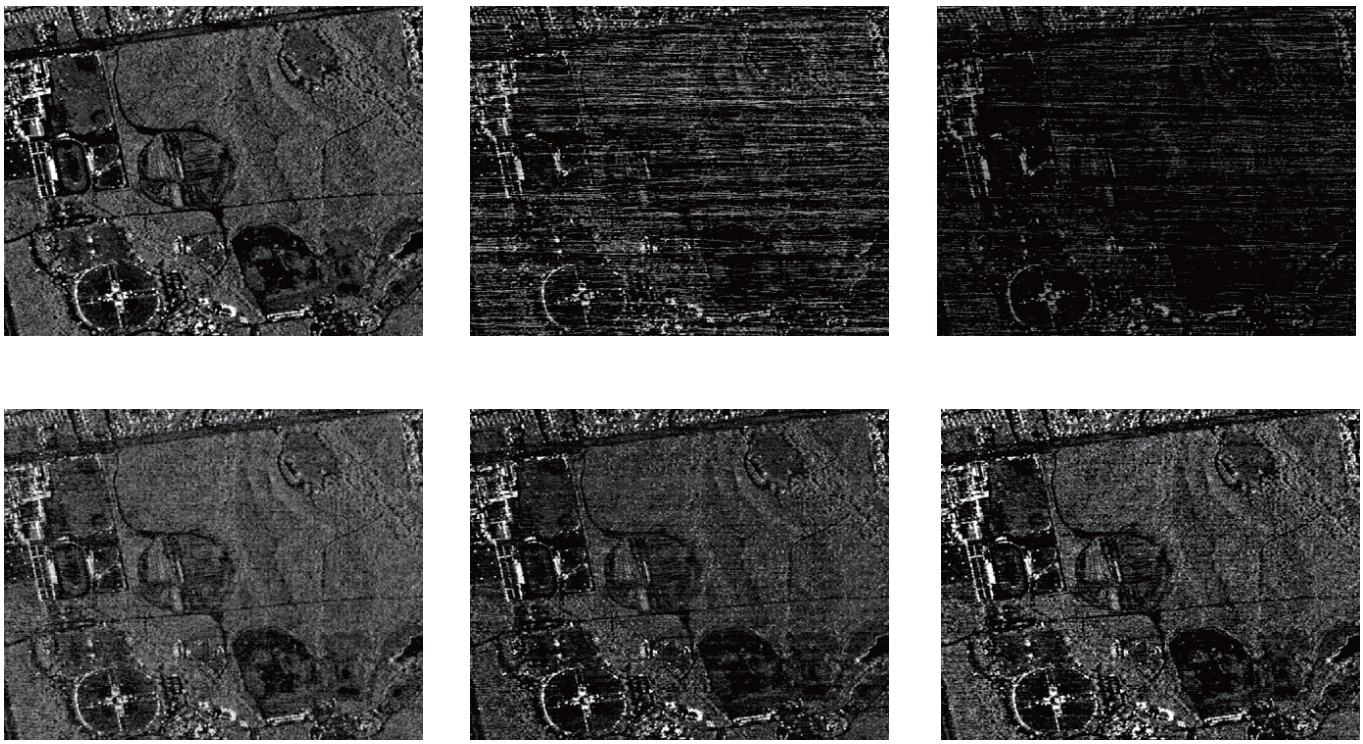

Fig. 5. Imaging results of scene 25 in experiment 1 obtained by different methods with 60% sampling rate. Columns 1 to 3 of the first row correspond to the original SAR image, sparse recovery method, and LFIST-Net, respectively. Columns 1 to 3 of the second row correspond to low-rank recovery method, joint low-rank and sparse recovery method, and the proposed method, respectively.

reconstruction performance can be observed using the joint low-rank and sparse property, including the iterative joint low-rank and sparse recovery method and the proposed LRSR-ADMM-Net, compared with other state-of-the-art recovery methods.

To quantitatively illustrate the benefits of the proposed method, we introduce normalized mean squared error (NMSE), peak signal-to-noise ratio (PSNR), and structural similarity index for measuring (SSIM) to quantitatively analyze the reconstruction quality of the images obtained by different methods. NMSE is defined as

$$\text{NMSE} = \frac{\|\Gamma_{\text{or}} - \Gamma_{\text{re}}\|_F^2}{\|\Gamma_{\text{or}}\|_F^2} \tag{31}$$

where Γ_{or} denotes the original reflectivity coefficients of the SAR image and Γ_{re} represents the recovered reflectivity coefficients of the SAR image.

The PSNR can be calculated by

$$\text{PSNR} = 20\log_{10}\left(\frac{\max(|\Gamma_{\text{or}}|)}{\text{RMSE}}\right). \tag{32}$$

Note that RMSE is the root mean square error of the recovered image and can be calculated by

$$\text{RMSE} = \sqrt{\frac{1}{MN}\|\Gamma_{\text{or}} - \Gamma_{\text{re}}\|_F^2}. \tag{33}$$

The definition of SSIM is given as

$$\text{SSIM}(\mathbf{x}, \mathbf{y}) = \frac{(2\mu_{\mathbf{x}}\mu_{\mathbf{y}} + c_1)(2\sigma_{\mathbf{xy}} + c_2)}{(\mu_{\mathbf{x}}^2 + \mu_{\mathbf{y}}^2 + c_1)(\sigma_{\mathbf{x}}^2 + \sigma_{\mathbf{y}}^2 + c_2)} \tag{34}$$

where $\mathbf{x}$ and $\mathbf{y}$ are the two images to be evaluated. $\mu_{\mathbf{x}}$ and $\mu_{\mathbf{y}}$ are the means of $\mathbf{x}$ and $\mathbf{y}$, respectively. $\sigma_{\mathbf{x}}$ and $\sigma_{\mathbf{y}}$ are the standard deviations of $\mathbf{x}$ and $\mathbf{y}$, respectively. $\sigma_{\mathbf{xy}}$ is the cross-covariance of the two images. c_1 and c_2 are constant which can be calculated by $c_1 = (k_1 L)^2$ and $c_2 = (k_1 L)^2$, respectively. Note that L is the dynamic range of the pixel values. Generally, the values of k_1 and k_2 are set to be 0.01 and 0.03, respectively.

The NMSE, PSNR, and SSIM values of the reconstructed images obtained by different methods are listed in Table III. It can be found that the conventional joint low-rank and sparse recovery method can achieve nearly 7.56 dB PSNR improvement for scene 3 with 60% sampling rate compared with the CS-based method. Although the LFIST-Net can improve the reconstruction performance by learning from the data, the reconstruction quality of the image is quite limited. As for the proposed LRSR-ADMM-Net, a better reconstruction performance compared with the iterative joint low-rank and sparse recovery method can be achieved. Note that the iterative parameters in the conventional joint low-rank and sparse recovery method are optimized after many manual adjustments. Nevertheless, the conventional joint low-rank and sparse recovery method still cannot achieve the same level of reconstruction performance as the LRSR-ADMM-Net. Meanwhile, it can be found in Table III that the reconstruction performance degrades with the decrease of the sampling rate for all the methods. Furthermore, we can observe that the reconstruction performance of the proposed LRSR-ADMM-Net, including NMSE, PSNR, and SSIM values, outperforms other state-of-the-art recovery methods for different subsampling rates.

To clearly show the impact of the iterative parameters of the joint low-rank and sparse recovery on the reconstruction

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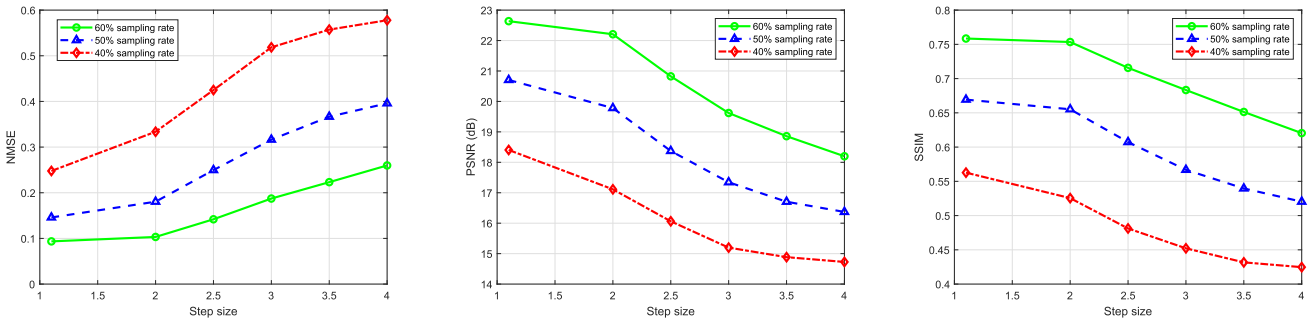
TABLE III
COMPARISONS OF NUMERICAL EVALUATIONS [NMSE, PSNR AND SSIM] OF DIFFERENT METHODS FOR EXPERIMENT 1

		60% sampling rate		50% sampling rate		40% sampling rate	
		scene 3	scene 25	scene 3	scene 25	scene 3	scene 25
CS-based method	NMSE	0.5329	0.6490	0.7349	0.8289	0.9613	1.0026
	PSNR	15.078	12.895	13.683	11.833	12.516	11.007
	SSIM	0.4368	0.3495	0.3447	0.2611	0.2672	0.1997
LFIST-Net	NMSE	0.2532	0.1869	0.3101	0.2225	0.3933	0.2404
	PSNR	18.310	18.301	17.430	17.545	16.397	17.209
	SSIM	0.4918	0.4509	0.4274	0.3856	0.3571	0.3360
MS-based method	NMSE	0.1831	0.0947	0.2624	0.1390	0.3593	0.2115
	PSNR	19.719	21.255	18.155	19.587	16.790	17.766
	SSIM	0.6627	0.7376	0.5725	0.6451	0.4833	0.5269
Joint recovery method	NMSE	0.0935	0.0871	0.1459	0.1369	0.2477	0.1931
	PSNR	22.637	21.618	20.704	19.653	18.405	18.160
	SSIM	0.7585	0.7385	0.6693	0.6387	0.5627	0.5467
LRSR-ADMM-Net	NMSE	0.0684	0.0538	0.1168	0.0885	0.1997	0.1276
	PSNR	23.994	23.710	21.670	21.551	19.341	19.960
	SSIM	0.8261	0.8235	0.7544	0.7561	0.6133	0.6479



(a) (b) (c)

Fig. 6. Achievable reconstruction performance of the joint low-rank and sparse recovery method with respect to step size. (a) NMSE. (b) PSNR. (c) SSIM.

performance, further numerical simulations are performed. The main iterative parameters used in the joint low-rank and sparse recovery method include the step size α and the weight parameter λ between the low-rank and sparse parts. Fig. 6 shows the reconstruction performance of scene 3 with respect to the step size α. It can be found that the reconstruction performance highly depends on the applied value of α. Note that the convergence of the joint low-rank and sparse method is determined by the step size. Less computation time is required with a larger step size. However, it can be found from Fig. 6 that the reconstruction performance degrades with the increase of the step size. Therefore, how to determine the value of the step size is a tricky issue in the joint low-rank and sparse recovery method. Besides, the influence of the weight parameter λ on the reconstruction performance is given in Fig. 7. Similarly, the value of the weight parameter λ has a significant impact on the reconstruction performance. The reconstructed performance suffers up to 7 dB PSNR value degradation when the improper value is applied. In practice, the optimal values of the iterative parameters are hard to set even though a large number of experiments has been performed. The high computational load and the problem of how to set proper iterative parameters in the iterative joint low-rank and sparse recovery method restrict its practical applications. On the other hand, the proper parameters can be obtained by the offline training in the proposed LRSR-ADMM-Net. After that, the well-trained network can be applied to obtain the final results with better reconstruction performance and higher computational efficiency.

TABLE IV
SYSTEM PARAMETERS OF EXPERIMENT 2

Parameter	Value
Carrier frequency	40 GHz
Signal bandwidth	800 MHz
Pulse width	1.5 μs
Fast time sampling rate	1000 MHz
Synthetic aperture time	2.90 s
Velocity of the platform	40 m/s
Slant range	1 km

B. Experiment 2

To further validate the performance of the proposed method, another experiment is conducted. The system parameters of the SAR system used in experiment 2 are provided in Table IV.

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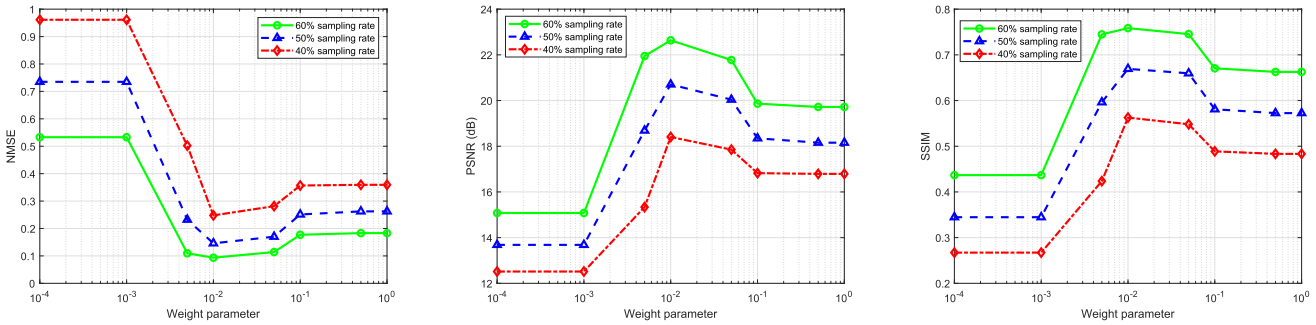

Fig. 7. Achievable reconstruction performance of the joint low-rank and sparse recovery method with respect to weight parameter. (a) NMSE. (b) PSNR. (c) SSIM.

TABLE V
COMPARISONS OF NUMERICAL EVALUATIONS [NMSE, PSNR, AND SSIM] OF DIFFERENT METHODS FOR EXPERIMENT 2

		60% sampling rate		50% sampling rate		40% sampling rate	
		scene 9	scene 17	scene 9	scene 17	scene 9	scene 17
CS-based method	NMSE	0.5518	0.5030	0.7056	0.7091	0.8669	0.9521
	PSNR	21.261	19.440	20.194	17.978	19.299	16.669
	SSIM	0.5639	0.6279	0.4763	0.5259	0.3948	0.4188
LFIST-Net	NMSE	0.2923	0.2999	0.3764	0.4424	0.4106	0.5836
	PSNR	24.020	21.685	22.922	19.998	22.545	18.794
	SSIM	0.5872	0.6258	0.5060	0.5283	0.4449	0.4272
MS-based method	NMSE	0.1929	0.4246	0.2770	0.5801	0.3806	0.7628
	PSNR	25.826	20.176	24.254	18.820	22.875	17.631
	SSIM	0.7488	0.6476	0.6640	0.5609	0.5714	0.4663
Joint recovery method	NMSE	0.1637	0.2519	0.2353	0.4015	0.3597	0.5921
	PSNR	26.538	22.443	24.963	20.418	23.119	18.731
	SSIM	0.7802	0.7409	0.6987	0.6406	0.5944	0.5354
LRSR-ADMM-Net	NMSE	0.0960	0.1503	0.1698	0.2863	0.2173	0.4085
	PSNR	28.854	24.685	26.380	21.887	25.308	20.343
	SSIM	0.8380	0.8261	0.7455	0.7306	0.6722	0.6469

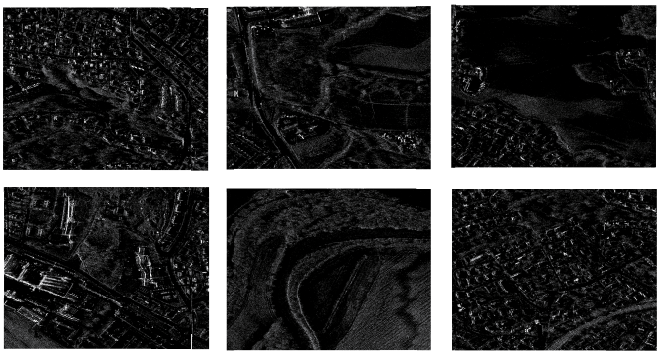

Fig. 8. Six randomly selected images from training dataset in experiment 2.

The training set consists of 400 SAR images which are also cropped from another SAR image by Airbus. The size of the cropped image is also 512 × 512. Fig. 8 shows some typical images which are randomly selected from the training dataset 2. We can observe the imaging scenes also include complex surface features, which are quite different from those in the training dataset 1. Similarly, the corresponding echoes of the scenes are generated with the amplitude of the image in the dataset using the system parameters given in Table IV.

The reconstructed results of different imaging scenes in experiment 2 obtained by different methods with the sub-sampled echo are shown in Figs. 9 and 10. It can be found from Fig. 9 that the correlation part plays a key role in imaging scene 9. Therefore, the method utilizing the low-rank property can obtain acceptable reconstruction performance which can also be observed from the imaging results obtained by the MS-based method. On the contrary, the method using the sparse property to recover the original image will incur performance degradation. Therefore, the CS-based method and the LFIST-Net cannot obtain good reconstruction performance in this case. Therefore, the imaging results of scene 9 obtained by the sparse-based methods aliased seriously, which can be observed from the second and third subimages in Fig. 9. As for the original SAR image of scene 17 provided in Fig. 10, the dominant scatters occupy most energy of the SAR image. Therefore, the CS-based and LFIST-Net methods can obtain acceptable recovered images in this case, which are shown in the second and third subimages of Fig. 10. The MS-based method suffers performance degradation in this situation which

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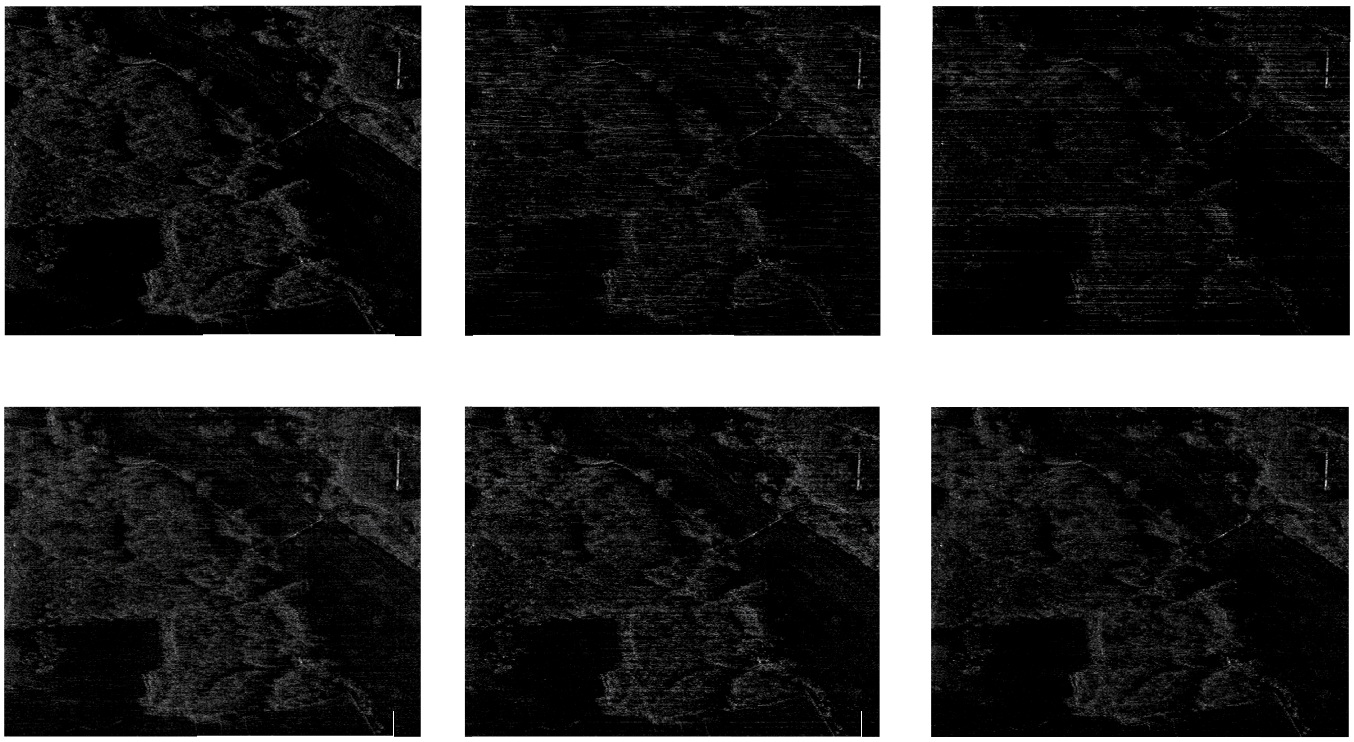

Fig. 9. Imaging results of scene 9 in experiment 2 obtained by different methods with 60% sampling rate. Columns 1 to 3 of the first row correspond to the original SAR image, Sparse recovery method, and LFIST-Net, respectively. Columns 1 to 3 of the second row correspond to low-rank recovery method, joint low-rank and sparse recovery method, and the proposed method, respectively.

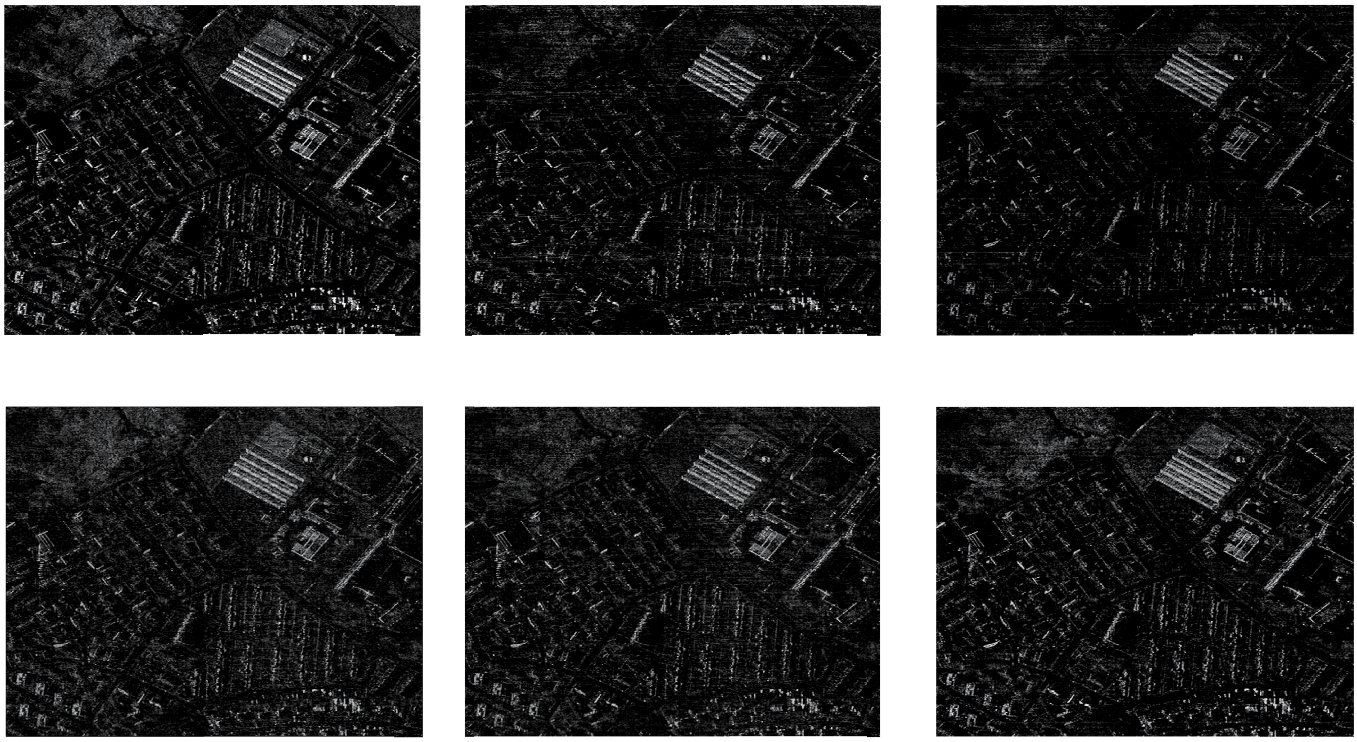

Fig. 10. Imaging results of scene 17 in experiment 2 obtained by different methods with 60% sampling rate. Columns 1 to 3 of the first row correspond to the original SAR image, Sparse recovery method, and LFIST-Net, respectively. Columns 1 to 3 of the second row correspond to low-rank recovery method, joint low-rank and sparse recovery method, and the proposed method, respectively.

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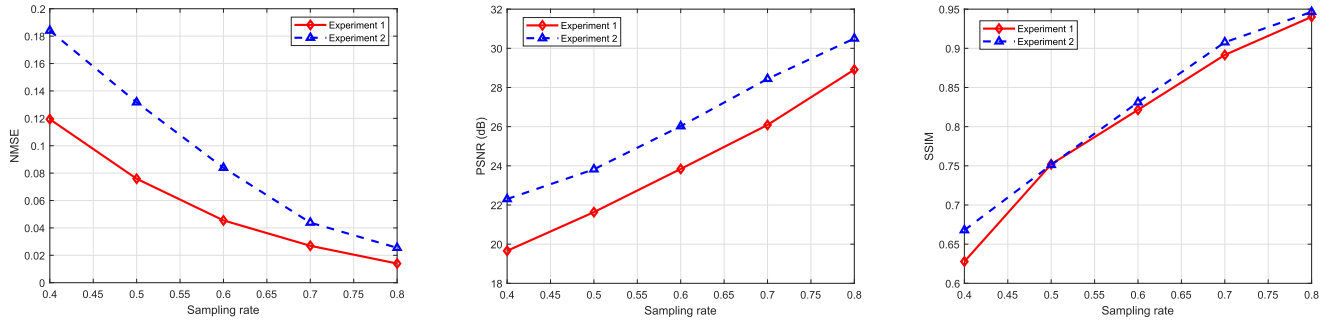
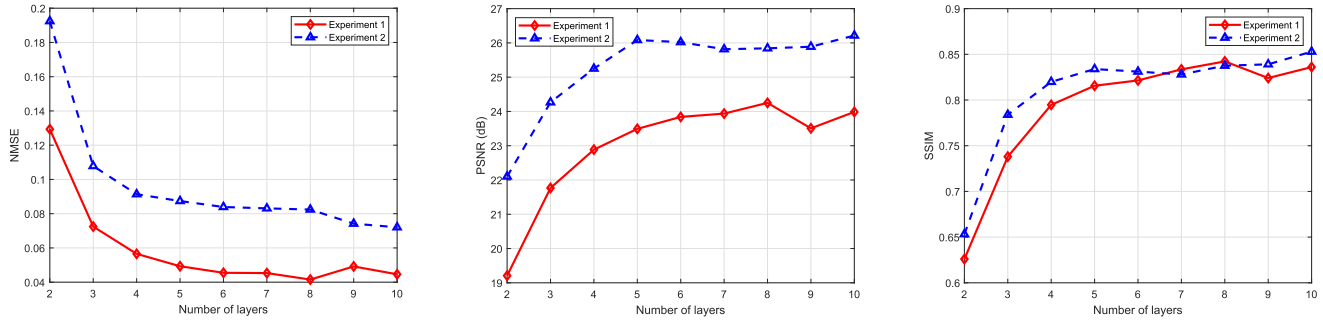


(a) (b) (c)

Fig. 11. Achievable reconstruction performance of LRSR-ADMM-Net with respect to sampling rate. (a) NMSE. (b) PSNR. (c) SSIM.



(a) (b) (c)

Fig. 12. Achievable reconstruction performance of LRSR-ADMM-Net with respect to number of updating layers. (a) NMSE. (b) PSNR. (c) SSIM.

is shown in the fourth subimage of Fig. 10. On the other hand, it can be observed from Figs. 9 and 10 that the joint low-rank and sparse recovery method and the proposed method can obtain good reconstructed images for the two imaging scenes, which shows the effectiveness of jointly utilizing the low-rank and sparse properties.

The NMSE, PSNR, and SSIM values of the reconstructed images in experiment 2 obtained by different methods are listed in Table V. It can be found that the MS-based method achieves a better reconstruction performance for scene 9 compared with sparse-based methods, which is consistent with the imaging results shown in Fig. 9. On the contrary, the sparse-based methods, i.e., the CS-based method and LFIST-Net, outperform the MS-based method for scene 17 reconstruction, which is also consistent with the previous analysis. Besides, the methods jointly using the low-rank and sparse properties surpass the methods utilizing the low-rank and sparse properties separately. Furthermore, the reconstruction performance of the proposed LRSR-ADMM-Net is better than the other state-of-the-art recovery methods for different subsampling rates, including the iterative joint low-rank and sparse recovery method.

C. Performance Analysis of LRSR-ADMM-Net

Fig. 11 gives the reconstruction performance, including NMSE, PSNR, and SSIM values, of the recovered image obtained by the proposed LRSR-ADMM-Net with respect to the subsampling rate. Note that the reconstruction performance is assessed by all the testing images where the testing set in experiment 1 contains 30 images and the testing set consists of 40 images in experiment 2. After performance evaluation for each testing image is completed, the average reconstruction performance is derived for experiment 1 and experiment 2, respectively. It can be found from Fig. 11 that the reconstruction performance improves when the subsampling rate increases.

Following that, we perform another performance analysis to evaluate the influence of the number of updating layers on the reconstruction performance. The achievable reconstruction performance with respect to the number of updating layers is shown in Fig. 12. We can observe that the reconstruction performance improves significantly with the increase of the applied updating layers when the number of layers is less than six for both experiments. This is because more learnable parameters and updates are introduced with the increase in the number of updating layers. On the other hand, it can be observed that the reconstruction performance enhances slightly when the number of layers is more than 6. It reveals that six updating layers are adequate to guarantee a satisfactory reconstruction performance, which means that the hundreds of iterations in the iterative joint low-rank and sparse recovery method can be replaced by the 6-layer LRSR-ADMM-Net with optimal parameters learned from the data.

VI. Conclusion

In this article, a nonsparse observation scene imaging method, i.e., LRSR-ADMM-Net, has been proposed. The architectures of the LRSR-ADMM-Net are derived from the

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iterative ADMM solver for joint sparse and low-rank matrices recovery. Meanwhile, the expressions of the trainable parameters in the LRSR-ADMM-Net are well-designed for faster convergence. The threshold parameters, and the weight parameter are learned automatically through end-to-end training. Experiments have been conducted to verify the effectiveness of the proposed method. The results reveal that the proposed LRSR-ADMM-Net outperforms the iterative recovery methods both in reconstruction performance and computational efficiency. Meanwhile, the proposed LRSR-ADMM-Net also shows its superiority in reconstruction performance in comparisons with the state-of-the-art trainable sparse-based network. Therefore, the proposed LRSR-ADMM-Net is capable of reconstructing the nonsparse observation scene with high efficiency and precision.

REFERENCES

[1] J. C. Curlander and R. N. McDonough, *Synthetic Aperture Radar*, vol. 11. New York, NY, USA: Wiley, 1991.

[2] Z. Li *et al.*, "Hybrid SAR-ISAR image formation via joint FrFT-WVD processing for BFSAR ship target high-resolution imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[3] T. Balz and M. Liao, "Building-damage detection using post-seismic high-resolution SAR satellite data," *Int. J. Remote Sens.*, vol. 31, no. 13, pp. 3369–3391, Jul. 2010.

[4] Z. Li *et al.*, "Bistatic SAR clutter-ridge matched STAP method for nonstationary clutter suppression," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[5] J. Zheng, T. Yang, H. Liu, and T. Su, "Efficient data transmission strategy for IIoTs with arbitrary geometrical array," *IEEE Trans. Ind. Informat.*, vol. 17, no. 5, pp. 3460–3468, May 2021.

[6] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.

[7] E. J. Candés, J. K. Romberg, and T. Tao, "Stable signal recovery from incomplete and inaccurate measurements," *Commun. Pure Appl. Math.*, vol. 59, no. 8, pp. 1207–1223, 2006.

[8] E. J. Candès and M. B. Wakin, "An introduction to compressive sampling," *IEEE Signal Process. Mag.*, vol. 25, no. 2, pp. 21–30, Mar. 2008.

[9] J. Zheng *et al.*, "An efficient strategy for accurate detection and localization of UAV swarms," *IEEE Internet Things J.*, vol. 8, no. 20, pp. 15372–15381, Oct. 2021.

[10] J. Fang, Z. Xu, B. Zhang, W. Hong, and Y. Wu, "Fast compressed sensing SAR imaging based on approximated observation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 352–363, Jan. 2014.

[11] X. Dong and Y. Zhang, "A novel compressive sensing algorithm for SAR imaging," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 2, pp. 708–720, Feb. 2014.

[12] H. An, J. Wu, K. C. Teh, Z. Sun, and J. Yang, "Nonambiguous image formation for low-earth-orbit SAR with geosynchronous illumination based on multireceiving and CAMP," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 348–362, Jan. 2021.

[13] H. Bi, G. Bi, B. Zhang, W. Hong, and Y. Wu, "From theory to application: Real-time sparse SAR imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2928–2936, Apr. 2020.

[14] J. Zheng, T. Yang, H. Liu, T. Su, and L. Wan, "Accurate detection and localization of unmanned aerial vehicle swarms-enabled mobile edge computing system," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 5059–5067, Jul. 2021.

[15] H. An, J. Wu, K. C. Teh, Z. Sun, and J. Yang, "Simultaneous moving and stationary target imaging for geosynchronous spaceborne-airborne bistatic SAR based on sparse separation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6722–6735, Aug. 2021.

[16] D. Yang, G. Liao, S. Zhu, X. Yang, and X. Zhang, "SAR imaging with undersampled data via matrix completion," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 9, pp. 1539–1543, Sep. 2014.

[17] E. Mason, I.-Y. Son, and B. Yazici, "Passive synthetic aperture radar imaging using low-rank matrix recovery methods," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 8, pp. 1570–1582, Dec. 2015.

[18] M. Fornasier, H. Rauhut, and R. Ward, "Low-rank matrix recovery via iteratively reweighted least squares minimization," *SIAM J. Optim.*, vol. 21, no. 4, pp. 1614–1640, 2011.

[19] B. Recht, M. Fazel, and P. A. Parrilo, "Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization," *SIAM Rev.*, vol. 52, no. 3, pp. 471–501, 2010.

[20] W. Pu and J. Wu, "OSRanP: A novel way for radar imaging utilizing joint sparsity and low-rankness," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 868–882, 2020.

[21] H. An, J. Wu, K. C. Teh, Z. Sun, and J. Yang, "Geosynchronous spaceborne–airborne bistatic SAR imaging based on fast low-rank and sparse matrices recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[22] H. An, J. Wu, K. C. Teh, Z. Sun, Z. Li, and J. Yang, "Joint low-rank and sparse tensors recovery for video synthetic aperture radar imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[23] J. Gao, B. Deng, Y. Qin, H. Wang, and X. Li, "Enhanced radar imaging using a complex-valued convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 35–39, Jan. 2019.

[24] C. Hu, L. Wang, D. Zhu, G. Zhang, and O. Loffeld, "FCNN-based ISAR sparse imaging exploiting gate units and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] Q. Cheng, A. A. Ihalage, Y. Liu, and Y. Hao, "Compressive sensing radar imaging with convolutional neural networks," *IEEE Access*, vol. 8, pp. 212917–212926, 2020.

[26] X. Huang, J. Ding, and Z. Xu, "Real-time super-resolution ISAR imaging using unsupervised learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[27] X. Wei *et al.*, "ISAR high-resolution imaging method with joint FISTA and VGGNet," *IEEE Access*, vol. 9, pp. 86685–86697, 2021.

[28] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proc. Int. Conf. Mach. Learn.*, 2010, pp. 399–406.

[29] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing," *IEEE Signal Process. Mag.*, vol. 38, no. 2, pp. 18–44, Mar. 2021.

[30] Y. Yang, J. Sun, H. Li, and Z. Xu, "ADMM-CSNet: A deep learning approach for image compressive sensing," *IEEE Trans. Pattern Anal. Mach. Intell.* vol. 42, no. 3, pp. 521–538, Mar. 2020.

[31] S. Wei, J. Liang, M. Wang, J. Shi, X. Zhang, and J. Ran, "AF-AMPNet: A deep learning approach for sparse aperture ISAR imaging and autofocusing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[32] R. Li, S. Zhang, C. Zhang, Y. Liu, and X. Li, "A computational efficient 2-D block-sparse ISAR imaging method based on PCSBL-GAMP-Net," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[33] M. Wang *et al.*, "TPSSI-Net: Fast and enhanced two-path iterative network for 3D SAR sparse imaging," *IEEE Trans. Image Process.*, vol. 30, pp. 7317–7332, 2021.

[34] M. Wang, S. Wei, J. Liang, S. Liu, J. Shi, and X. Zhang, "Light-weight FISTA-inspired sparse reconstruction network for mmW 3-D holography," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[35] M. Wang *et al.*, "RMIST-Net: Joint range migration and sparse reconstruction network for 3-D mmW imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[36] M. Li *et al.*, "Target-oriented SAR imaging for SCR improvement via deep MF-ADMM-Net," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[37] M. Li, J. Wu, W. Huo, Z. Li, J. Yang, and H. Li, "STLS-LADMM-Net: A deep network for SAR autofocus imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[38] E. Mason, B. Yonel, and B. Yazici, "Deep learning for radar," in *Proc. IEEE Radar Conf. (RadarConf)*, May 2017, pp. 1703–1708.

[39] E. Mason, B. Yonel, and B. Yazici, "Deep learning for SAR image formation," *Proc. SPIE*, vol. 10201, Apr. 2017, Art. no. 1020104.

[40] Y. Wei, Y. Li, Z. Ding, Y. Wang, T. Zeng, and T. Long, "SAR parametric super-resolution image reconstruction methods based on ADMM and deep neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10197–10212, Dec. 2021.

[41] L. Kang, T. Sun, Y. Luo, J. Ni, and Q. Zhang, "SAR imaging based on deep unfolded network with approximated observation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[42] K. Aberman and Y. C. Eldar, "Sub-Nyquist SAR via Fourier domain range-Doppler processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6228–6244, Nov. 2017.

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[43] J. A. Tropp and A. C. Gilbert, "Signal recovery from random measurements via orthogonal matching pursuit," *IEEE Trans. Inf. Theory*, vol. 53, no. 12, pp. 4655–4666, Dec. 2007.
[44] D. L. Donoho and J. Tanner, "Thresholds for the recovery of sparse solutions via L_1 minimization," in *Proc. 40th Annu. Conf. Inf. Sci. Syst.*, Mar. 2006, pp. 202–206.
[45] M. Çetin *et al.*, "Sparsity-driven synthetic aperture radar imaging: Reconstruction, autofocusing, moving targets, and compressed sensing," *IEEE Signal Process. Mag.*, vol. 31, no. 4, pp. 27–40, Jul. 2014.
[46] L. C. Potter, E. Ertin, J. T. Parker, and M. Cetin, "Sparsity and compressed sensing in radar imaging," *Proc. IEEE*, vol. 98, no. 6, pp. 1006–1020, Jun. 2010.
[47] Z. Wang, M.-J. Lai, Z. Lu, W. Fan, H. Davulcu, and J. Ye, "Orthogonal rank-one matrix pursuit for low rank matrix completion," *SIAM J. Sci. Comput.*, vol. 37, no. 1, pp. A488–A514, Jan. 2015.
[48] Y. Hu, D. Zhang, J. Ye, X. Li, and X. He, "Fast and accurate matrix completion via truncated nuclear norm regularization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 9, pp. 2117–2130, Sep. 2013.
[49] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

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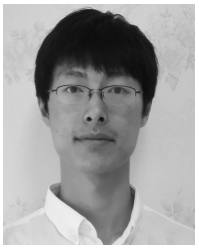

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